

Filippos N. Tzortzoglou

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EDUCATION

Cornell University — Ithaca, NY, USA

August 2023 - Present

3rd year PhD candidate in Civil and Environmental Engineering

Supervisor: Prof. Andreas A. Malikopoulos

Cornell University - Ithaca, NY, USA

August 2023 - November 2025

Master of Science in Transportation Engineering

Related Coursework:

Bayesian Estimation | Introduction to Reinforcement Learning | Optimal Control and Decision Theory | Information Design and Strategic Decisions | Foundations of Complex Systems | Transportation Energy Systems | Sustainable Transportation Systems Design | Preparing to Write Workshop

University of Delaware — Newark, DE, USA

August 2022 - August 2023

1st year PhD in Mechanical Engineering (continued Ph.D. studies at Cornell University)

Supervisor: Prof. Andreas A. Malikopoulos

Related Coursework:

Convex Optimization | Linear Systems | Statistical Computing and Optimization | Engineering Mathematics | Introduction to Numerical Discretization | Intermediate Dynamics

Technical University of Crete — Chania, Greece

September 2017 - July 2022

Diploma-Integrated Master's Degree in Production Engineering and Management

Related Coursework:

Programming in C | Statistics | Non-Linear Programming | Artificial Intelligence | Robotics | Mechatronics | Control Systems

HONORS AND AWARDS

- **IEEE Outstanding Student Paper Award 2025 on Smart Cities** for the paper: **F.N. Tzortzoglou**, L.E. Beaver, A. Malikopoulos, "A Feasibility Analysis at Signal-Free Intersections", IEEE Control Systems Letters (LCSS). [\[Award page\]](#)
- **University Wide Award for Excellence in Teaching** | **Category: Pedagogy**, Cornell University
Selected by the Graduate and Professional Student Assembly (GPSA) Faculty Advising, Teaching, Mentorship, and Awards Committee (FATMAC) from nominations spanning 10,000+ graduate and professional students. [\[Certificate\]](#)
- Selected as the **Student Spotlight** in Cornell University's weekly news (February 2026). [\[Feature article\]](#)
- **Finalist, Best Paper Award**, 17th IFAC Symposium on Control of Transportation Systems (CTS 2024), for the paper by H. Bang, A. Dave, **F.N. Tzortzoglou**, and A.A. Malikopoulos, "A Mobility Equity Metric for Multi-Modal Intelligent Transportation Systems". [\[Proceedings\]](#)
- NSF grant for *Non-Academic Research Internships for Graduate Students (INTERN) Program*.
- Fellowship from the Gerondelis Foundation.

PUBLICATIONS (JOURNAL PAPERS)

- **F.N. Tzortzoglou**, and L.E. Beaver. “[Mixed Traffic: A Perspective from Long Duration Autonomy](#)”, IEEE Transactions on Intelligent Transportation Systems (In Review).
- **F.N. Tzortzoglou**, and A.A. Malikopoulos. “[Teaching Cars to Drive: Toward Connected and Automated Vehicles](#)”, IEEE Potentials (In Press).
- **F.N. Tzortzoglou**, L.E. Beaver, A. Malikopoulos. “[A Feasibility Analysis at Signal-Free Intersections](#)”, IEEE Control Systems Letters (LCSS) (**IEEE Outstanding Student Paper Award 2025 on Smart Cities**).
- **F.N. Tzortzoglou**, Craig B., Meng X., and Malikopoulos, A. A. (2024). “[Handling Pedestrian Uncertainty in Coordinating Autonomous Vehicles at Signal-Free Intersections](#)”, Automatica (In Review).
- H. Bang, A. Dave, **F. N. Tzortzoglou**, A. Malikopoulos, “[On Mobility Equity and the Promise of Emerging Transportation Systems](#)”, IEEE Transactions on Intelligent Transportation Systems (T-ITS).
- V. Le, B. Chalaki, **F. N. Tzortzoglou**, A. A. Malikopoulos. “[Stochastic Time-Optimal Trajectory Planning for Connected and Automated Vehicles in Mixed-Traffic Merging Scenarios](#)”, IEEE Transactions on Control Systems Technology (TCST).

PUBLICATIONS (CONFERENCE PAPERS)

- **Filippos N. Tzortzoglou**, Clayton Walters, and Andreas A. Malikopoulos. “[Data-Driven Modeling and Prediction of Human Drivers in Mixed Traffic Using Virtual Reality](#)”, (Accepted in European Conference of Control 2026 (ECC)).
- **Filippos N. Tzortzoglou**, Logan E. Beaver, and Andreas A. Malikopoulos. “[Safe and Efficient Coexistence of Autonomous Vehicles with Human-Driven Traffic at Signalized Intersections](#)”, Proceedings of the 64th Conference on Decision and Control (CDC), IEEE, 2025.
- H. Bang, A. Dave, **F.N. Tzortzoglou**, and A.A. Malikopoulos, “[A Mobility Equity Metric for Multi-Modal Intelligent Transportation Systems](#)”, Proceedings of the 17th IFAC Symposium on Control in Transportation Systems, 2024 (**Finalist for best paper award**).
- **F.N. Tzortzoglou**, L.E. Beaver, and A. Malikopoulos, “[A Feasibility Analysis at Signal-Free Intersections](#)”, Proceedings of the 63rd Conference on Decision and Control (CDC), 2024 (also published in IEEE Control Systems Letters (LCSS)).
- **F.N. Tzortzoglou** and A.A. Malikopoulos, “[Signal-Free Intersections and Automated Vehicles — A Case Study in Heraklion, Crete, Greece](#)”, Conference on Emerging Technologies in Transportation Systems (TRC-30), 2024.
- **F.N. Tzortzoglou**, D. Theodosis, A. Dave, and A.A. Malikopoulos, “[Potential-Based Controller for Efficient Flow of Connected and Automated Vehicles](#)”, Proceedings of the 2024 American Control Conference (ACC), IEEE, 2024.
- **F.N. Tzortzoglou**, D. Theodosis, and A. Malikopoulos, “[An approach for optimizing acceleration in connected and automated vehicles](#)”, Transportation Research Board 103rd Annual Meeting (TRB), 2024.
- D. Theodosis, **F.N. Tzortzoglou**, I. Karafyllis, I. Papamichail, and M. Papageorgiou, “[Sampled-data controllers for autonomous vehicles on lane-free roads](#)”, Proceeding of the 30th Mediterranean Conference on Control and Automation (MED), IEEE, 2022.

INDUSTRY EXPERIENCE

Intern at MathWorks — Boston, MA, USA

June 2024 - August 2024

- Developed algorithms for the operation of Connected and Automated Vehicles with the presence of pedestrians in RoadRunner Scenario software.
- Utilized MathWorks tools (cameras, radars, lidars) for the detection and classification of objects in real-time through Simulink and apply efficient control strategies for the avoidance of such objects.
- Utilized Reinforcement Learning Toolbox developed by MathWorks.

RESEARCH EXPERIENCE

Research Assistant — Ithaca NY, USA

August 2023 - Present

Research Assistant in the Information and Decision Science Laboratory - Cornell University

- Involved in two NSF projects under Grants CNS-2149520 and CMMI-2219761. Principal Investigator: Andreas Malikopoulos.
- Project leader of a 3-member team working on Virtual Reality environments (Unity) in Mixed Traffic Scenarios.
- Trained members of the IDS Lab in PTV VISSIM software.
- Designed a simulation environment using PTV VISSIM software for the validation of efficient control strategies for Connected and Automated Vehicles operating in a mixed-traffic environment.
- Conducted simulations in MATLAB for the validation of control strategies associated with mobility equity and fair accessibility in services among neighborhoods in the Boston Metropolitan Area.
- Explored controller's feasibility for the operation of Connected and Automated Vehicles in signal-free intersections.

Research Assistant — Newark DE, USA

August 2022 - August 2023

Research Assistant in the Information and Decision Science Laboratory - University of Delaware

- Involved in two NSF projects under Grants CNS-2149520 and CMMI-2219761. Principal Investigator: Andreas Malikopoulos.
- Designed and improved controller strategies for the operation of Connected and Automated vehicles on lane-based road environments.

Research Assistant — Chania, Greece

July 2021 - July 2022

Research Assistant in the Dynamic Systems and Simulation Laboratory (DSSL Lab) - Technical University of Crete

- Involved in the ERC advanced grant with title: TrafficFluid — Lane-free Artificial-Fluid Environment for Vehicular Traffic. Principal Investigator: Markos Papageorgiou.
- Designed decentralized control strategies for the two-dimensional movement of autonomous vehicles on lane-free roads.
- Performed simulations using Mathematica' and MATLAB' softwares to visualize the related results.

Team Manager, Chania, Greece

August 2020 - January 2022

Team Manager of the TUCer team (**T**echnical **U**niversity of **C**rete **e**co **r**acing)

- Developed zero emission, low consumption, hydrogen-powered, electric prototype vehicles. TUCer team has won several awards at international low-consumption competitions.

TEACHING EXPERIENCE

Teaching Assistant — Newark, DE, USA

Aug 2022 – May 2023

- Teaching assistant at the University of Delaware, Mechanical Engineering Department, for Control Systems Laboratory and Solid Mechanics Laboratory.
- Held office hours.
- Graded assignments and exams.

Teaching Assistant — Ithaca, NY, USA

Aug 2025 – December 2025

- Teaching assistant at Cornell University for the course *Uncertainty for Engineers*.

SKILLS

- **Traffic Simulation Software:** Proficient in PTV VISSIM, SUMO, and RoadRunner. See illustrative projects in [VISSIM](#), [SUMO](#), [RoadRunner](#), [Automated Driving Toolbox \(by MATLAB\)](#).
- **Programming Languages:** Advanced in Python and Matlab; Intermediate in C. Over 4 years of experience in developing simulation and optimization algorithms.
- **Experience with GitHub:** See illustrative project at this [link](#).
- **Optimization Software:** Experienced in solving large-scale optimization problems using Gurobi software.
- **Machine Learning:** Knowledgeable in designing and implementing Neural Networks for data analysis and predictive modeling.